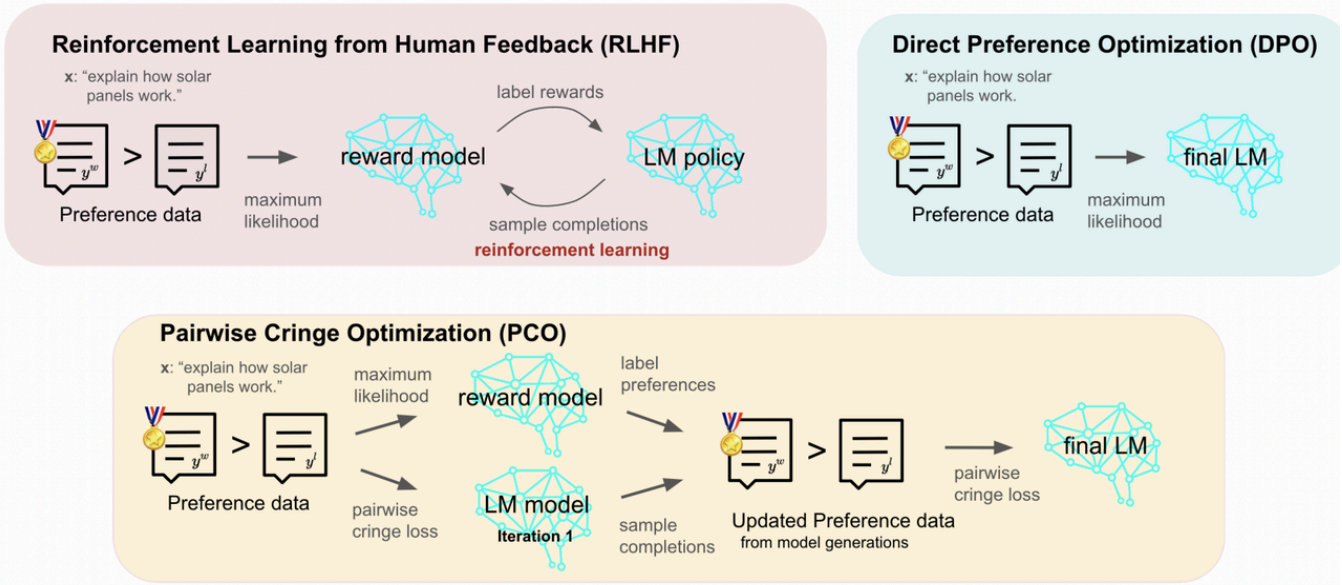
Some things are more CRINGE than others: Iterative Preference Optimization with the Pairwise Cringe Loss

> #Finetuning #LossFunction



- What problems does this framework solve?
  - The commonly used data type in alignment is preference-based (A is better than B).
  - Normally used DPO and PPO are sequence-level optimization, and cringe loss, a token-level optimization method, works well for score-based datasets only (**too hard** for preference, and lack a mechanism to control the strength when the diff between good answer and the bad answer is big enough).
- What does this framework do?
  - [Background] Cringe loss:
    - When learning a good response, use CE loss:

      $$\mathcal{L}_{CE}(x^+, y^+) = -\log p([x^+, y^+]) = -\log p(x^+) - \log p(y^+|x^+)$$
    - When learning from a bad response, instead of decreasing the total probability of its generation, we correct the "bad tokens" with "good tokens". To find out the "good" token, we select k tokens with the top logits values (with the original token removed), re-calculate its softmax, and sample a word from the k tokens.
    - Use contrasive loss, maximize the possibility of the good token selected:

      $$\mathcal{L}_{Cr}(x^-, y^-) = -\log \frac{\exp(s_t^*)}{\exp(s_t^*)+\exp(s_t[y_t^-])}$$
    - Add them together:

      $$\mathcal{L}_{Bin}(x, y^+, x, y^-) = \mathcal{L}_{CE}(x, y^+) + \alpha\mathcal{L}_{Cr}(x, y^-)$$
  - Apply to pairwise preference:
    - We firstly find out the difference in probability of generating the +-pairs:

$M(x, y^w, y^l) = \log p(y^w|x) - \log p(y^l|x)$

- We only want to apply cringe loss when $M < 0$ (more likely to generate the bad answer).

- Use M as a gated value:

$g(x, y^w, y^l) = \sigma\left(\frac{b - M(x, y^w, y^l)}{\tau}\right)$

- When $M < b$, $g \to 1$; otherwise $g \to 0$. b is the margin to control the strength of the loss.

- Finally, multiply the gate to the cringe loss:

$\mathcal{L}_{Pair}(x, y^w, y^l) = g(x, y^w, y^l)\mathcal{L}_{Bin}(x, y^w, x, y^l)$

- Can be used iteratively.

- Experiment Results
  - Reducing Generated Redundancy: In a controlled experiment designed to reduce textual repetition, Pairwise Cringe successfully lowered the repetition rate, outperforming the baseline Binary Cringe and achieving results comparable to DPO—even slightly surpassing it in generative quality (F1 score).
  - AlpacaFarm Comprehensive Instruction Fine-Tuning: On the Benchmark for general instruction following, Pairwise Cringe achieved a win rate of 52.0% after the first training round, surpassing DPO (50.2%) and PPO (48.5%). Following a second iteration (PCO), the win rate further increased to 54.7%.
  - Soft vs. Hard Margin Comparison: Experiments demonstrate that gradient-based soft margins significantly outperform rigid hard margins.
- Work Cited
  - Paper
  - NotebookLM
  - Chat History